

SQL JOIN Implementation

1) INNER JOIN

Employees with department name and projects they work on

The screenshot shows a SQL Query Builder interface with a 'Worksheet' tab and a 'Query Builder' tab. The query is as follows:

```
SELECT
    e.SSN,
    e.Fname || ' ' || e.Lname AS Employee_Name,
    d.DName AS Department_Name,
    p.PName AS Project_Name
FROM Employee e
INNER JOIN Department d
    ON e.Dnum = d.Dnum
INNER JOIN Works_On w
    ON e.SSN = w.SSN
INNER JOIN Project p
    ON w.PNumber = p.PNumber;
```

Below the query, the 'Query Result' tab is active, showing the results of the query. The status bar indicates 'All Rows Fetched: 5 in 0.023 seconds'. The results are displayed in a table with the following columns: SSN, EMPLOYEE_NAME, DEPARTMENT_NAME, and PROJECT_NAME.

	SSN	EMPLOYEE_NAME	DEPARTMENT_NAME	PROJECT_NAME
1	111	Ali Ahmed	IT	Website System
2	222	Sara Khalid	IT	Website System
3	222	Sara Khalid	IT	HR Portal
4	333	Omar Said	HR	HR Portal
5	444	Maha Nasser	Finance	Finance App

2) LEFT JOIN

All employees with department and dependent details

Worksheet Query Builder

```

SELECT
  e.SSN,
  e.Fname || ' ' || e.Lname AS Employee_Name,
  d.DName AS Department_Name,
  dep.DependentName,
  dep.Gender AS Dependent_Gender,
  dep.BirthDate AS Dependent_BirthDate
FROM Employee e
LEFT JOIN Department d
  ON e.Dnum = d.Dnum
LEFT JOIN Dependent dep
  ON e.SSN = dep.SSN;

```

Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.005 seconds

	SSN	EMPLOYEE_NAME	DEPARTMENT_NAME	DEPENDENTNAME	DEPENDENT_GENDER	DEPENDENT_BIRTHDATE
1	111	Ali Ahmed	IT	Hamad	Male	10-APR-15
2	111	Ali Ahmed	IT	Aisha	Female	12-SEP-18
3	222	Sara Khalid	IT	(null)	(null)	(null)
4	333	Omar Said	HR	Salim	Male	20-JUN-16
5	444	Maha Nasser	Finance	(null)	(null)	(null)

3) RIGHT JOIN

All projects with department and employees, including projects with no employees

Worksheet

Query Builder

```

SELECT
    p.PNumber,
    p.PName AS Project_Name,
    d.DName AS Department_Name,
    e.Fname || ' ' || e.Lname AS Employee_Name
FROM Employee e
INNER JOIN Works_On w
    ON e.SSN = w.SSN
RIGHT JOIN Project p
    ON w.PNumber = p.PNumber
INNER JOIN Department d
    ON p.Dnum = d.Dnum;

```

Script Output x

Query Result x

SQL | All Rows Fetched: 5 in 0.01 seconds

	PNUMBER	PROJECT_NAME	DEPARTMENT_NAME	EMPLOYEE_NAME
1	100	Website System	IT	Ali Ahmed
2	100	Website System	IT	Sara Khalid
3	200	HR Portal	HR	Sara Khalid
4	200	HR Portal	HR	Omar Said
5	300	Finance App	Finance	Maha Nasser

4) SELF JOIN

Each employee with supervisor name and department

```
SELECT
  e.SSN,
  e.Fname || ' ' || e.Lname AS Employee_Name,
  s.Fname || ' ' || s.Lname AS Supervisor_Name,
  d.DName AS Department_Name
FROM Employee e
LEFT JOIN Employee s
  ON e.SupervisorSSN = s.SSN
LEFT JOIN Department d
  ON e.Dnum = d.Dnum;
```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.005 seconds

	SSN	EMPLOYEE_NAME	SUPERVISOR_NAME	DEPARTMENT_NAME
1	222	Sara Khalid	Ali Ahmed	IT
2	333	Omar Said	Ali Ahmed	HR
3	444	Maha Nasser	Omar Said	Finance
4	111	Ali Ahmed		IT